

CSE4227 Digital Image Processing

Introductory Class

Dr. Kazi A Kalpoma

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& Director of ICT

Ahsanullah University of Science & Technology (AUST)

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Google Class code: cykjms3

WELCOME TO MY CLASS



kalpoma@aust.edu

Course Teacher's Details

- ❑ **B.Sc.:** Applied Physics, Electronics and Communication Engineering, Dhaka University. 1994
- ❑ **M.Sc.:** Computer Science and Engineering, Dhaka University. 1996
- ❑ **M.Sc.:** Information Sciences, Dept. of Computer and Mathematical Sciences, Tohoku University, Japan. 1998-2000
- ❑ **PhD.:** Information Sciences, Dept. of Computer and Mathematical Sciences, Tohoku University, Japan 2004-2007.
- ❑ **Post Doctoral as a Research Fellow** at Tohoku University, Japan 2009~2012.

Continuation....

- ☐ **Lecturer / Assistant / Associate / Professor**
at AIUB, May 2001 ~ Sept 2017
- ☐ **Visiting Professor** at Tohoku University, April-June 2017
- ☐ **Professor** at AUST, Nov 2017 ~ date
- ☐ **Director** of ICT, AUST May 2019 ~ date
- ☐ **IEEE Senior Member**
- ☐ Reviewer, IEEE(more than a decade.), TCM, PCM of
different International Conferences
- ☐ [My Profile](#)

The list of courses I taught:

1. Digital Image Processing (M.Sc./B.Sc.)
2. Computer Vision and Pattern Recognition (M.Sc.)
3. Machine Learning and Neural Network (M.Sc.)
4. Computer Organization and Architecture (M.Sc./B.Sc.)
5. Advanced Operating System (M.Sc.)
6. Operating System
7. Programming Language (C/C++)
8. Discrete Mathematics
9. Data Structure
10. Research Methodology

My Research Area:

- 1. Image processing**
- 2. Satellite Image Processing and Its Applications**
- 3. Computer Vision and Pattern Recognition**
4. Human Computer Interaction
5. Artificial Intelligence and Machine Learning
6. Cyber Security

Contact Information

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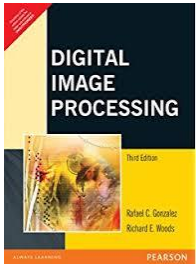
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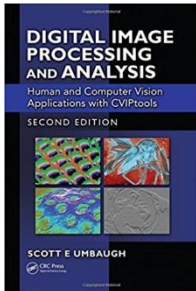
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Reading Materials

Textbook



R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008



Scott E Umbaugh, Digital Image Processing and Analysis: Human and Computer Vision Applications with CVPTools, Second Edition

Reading Materials

- R.C. Gonzalez, R.E. Woods, and S.L. Eddins, Digital Image Processing Using MATLAB (2nd Edition), 2009
- M. Sonka, V. Hlavac, and R. Boyle, Image Processing, Analysis, and Machine Vision, Thomson, 2008
- “Fundamental of Digital Image Processing”, Anil K. Jain.
- Anil K. Jain, , Fundamentals of Digital Image Processing', Pearson 2002.
- Kenneth R. Castleman, Digital Image Processing, Pearson, 2006.
- D,E. Dudgeon and RM. Mersereau, , Multidimensional Digital Signal Processing', Prentice Hall Professional Technical Reference, 1990.
- ✓ My Slides
- ✓ <https://www.tutorialspoint.com/dip/index.htm>

Marks Distribution:

Attendance and Class Performance	10
Class Test	20
Final Exam	70
Total	100

Four class tests will be taken and best THREE will be considered for “Class Test” marks.

PRE-REQUEST:

Students will be expected to be familiar with

☐ Data Structure

☐ Basic Probability/Statistics

☐ Linear Algebra

☐ Vector Calculus

Course Contents

- ❑ **Digital Image Fundamentals** (Chap-1&2)
 - Digital image processing (DIP)
 - Key stages of DIP
 - Digital image Representation
- ❑ **Image Sensing and Acquisition** (Chap-2)
 - Human Perception, Image Formation, Image Acquisition
 - Image Digitization (sampling and quantization)
 - Pixels & their relationships
- ❑ **Intensity Transformations and Spatial Filtering** (Chap-3&4)
 - Image enhancement in **spatial domain**
 - Histogram processing
 - Image enhancement in **frequency domain**
- ❑ **Morphological Image Processing** (Chap-9)
 - Erosion, Dilation, Opening and Closing, Hit-or-Miss Transformation, Thinning, Thickening, Skeletons.
- ❑ **Image Segmentation** (Chap-10)
 - Edge/Line detection
 - Segmentation
- ❑ **Image Compression** (Chap-8)
 - ❑ Data Redundancy,
 - ❑ Image Entropy,
 - ❑ Different Compression Techniques
- ❑ **Color Image processing fundamentals** (Chap-6)
 - Color Fundamentals
 - Color Models and Examples
- ❑ **Image Restoration** (Chap-5)